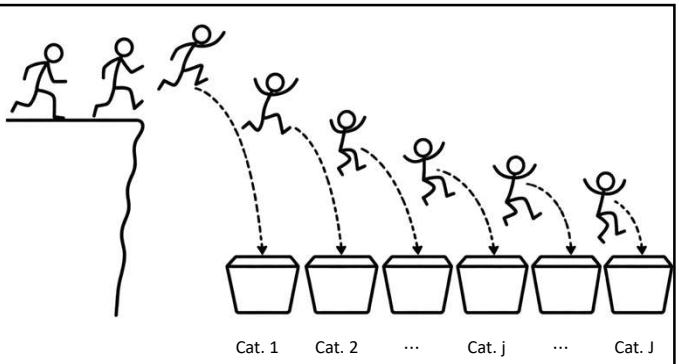
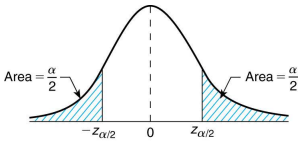
Agenda

- Earlier
  - Qualitative variables
  - Hypothesis testing: Table of counts, Type II errors
- Today
  - Goodness-of-fit when probabilities are not fully specified
  - Poisson
  - Multivariate Gaussian





1

Review

Dice 210 Rolls							
Outcome	1	2	3	4	5	6	Total
Count	26	40	37	26	43	38	210
Proportion	0.124	0.190	0.176	0.124	0.205	0.181	1.000

- Are these the rolls of a fair dice or not?
  - Null hypothesis H_0 with equal probabilities of 1/6
  - Fair dice: Count of 35 for each case
  - chi-square distribution, df = 5
  - $x_0^2 = \frac{(26-35)^2}{35} + \cdots + \frac{(38-35)^2}{35} = 7.54$
  - p-value = 0.18

Intuition: Degree of freedom lost because $\Sigma p_i = 1$

```
octave:1> chi2cdf (7.54, 5)
ans = 0.8165
octave:2> 1- chi2cdf (7.54, 5)
ans = 0.1835
octave:3>
```

- We do not have any evidence to reject the null hypothesis (i.e., we fail to reject the null hypothesis)

$$x_o^2 = \Sigma \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

2

Review: Goodness of fit

- n independent variables $Y_i, \dots, Y_n$ each taking one of the values $1,2, \dots, k$
- Null hypothesis $H_0: p_i: i = 1, \dots, k$ is the pmf
- Alternate hypothesis $H_a: P(Y = i) \neq p_i$ for some $i = 1, \dots, k$
- Let $X_i, (i = 1, .., k)$ be the number of the Y_i that equal i
- Under $H_0, X_i \sim \text{Bin}(n, p_i)$
  - statistic $T = \Sigma_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i}$
  - When n is large, if H_o is true, then $T \sim \chi^2(k-1)$
  - Reject H_o using p-value as usual

Intuition: Don't want numerator to be large compared to the denominator

Intuition: Degree of freedom lost because $\Sigma X_i = n$

3

Example

Dice 210 Rolls							
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Count	26	40	37	26	43	38	210
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- Are these the rolls of a fair dice or not?
  - Null hypothesis H_0 with equal probabilities of 1/6
  - Fair dice: Count of 35 for each case
  - df = 5
  - $x_0^2 = \frac{(26-35)^2}{35} + \cdots + \frac{(38-35)^2}{35} = 7.54$
  - P-value = 0.18
- We do not have any evidence to reject the null hypothesis (i.e., we fail to reject the null hypothesis)
- But the dice could be still be biased and we may be making a type-II error

Example: If there was a manufacturing defect, and the true probabilities was (0.25, 0.15, 0.15, 0.15, 0.15, 0.15)

4

Review: Type I and Type II errors

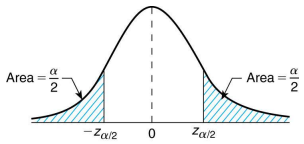
- There can be two types of errors
  - Type I: Incorrectly recommends rejection of H_0 (even if it is true)
  - Type II: Incorrectly fail to reject H_0 (even if it is false)
- Ideally, H_0 should be rejected only if the samples are very unlikely when H_0 is true
  - This calls for increasing sample size

Decision Making		
	Actual situation	
Decision made	H_0 is true	H_0 is false
Do not reject H0	OK	Type II error
Reject H_0 as false	Type I error	OK

The power of a test = 1 - probability of type II error

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Example



- 5 signals are received by a transmitter on a particular day with the mean value of $\bar{X} = 9.5$ (signal could be corrupted)
  - Null hypothesis: signal is guessed to have $\mu_0 = 8$, noise is $\mathcal{N}(0,4)$, true mean is 10, not 8
  - Test statistic $t_0 = (9.5 - 8)/\text{se}(\bar{x}) = 1.68$, 1.68 < 1.96, do not reject
  - $(\mu_0 - \mu)/(2/\sqrt{5}) = -\sqrt{5}$
  - $\beta = \phi(-\sqrt{5} + 1.96) - \phi(-\sqrt{5} - 1.96) = 0.392$
- The null hypothesis is false, the probability of not rejecting is 39%
- Power of this method = 0.61

$$\beta(\mu) = \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)$$

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Type II error: Chi-Square Goodness of Fit

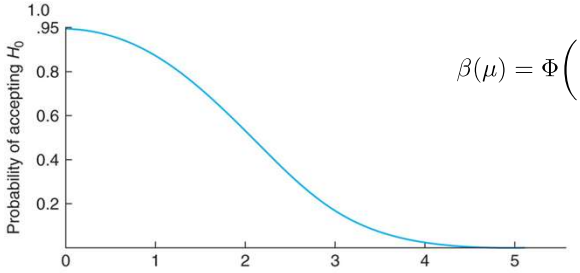
- $H_0: \mathbf{p} = \mathbf{p}_0 = (1/6,1/6,1/6,1/6,1/6,1/6)$.
- What is the probability of failing to reject the null hypothesis H_0 at a significance level α when the true mean probability vector p is unequal to p_0?
- Define $\beta(\mathrm{p})$ to be the probability of not rejecting H_0

Example: If there was a manufacturing defect, and the true probabilities was (0.25, 0.15, 0.15, 0.15, 0.15, 0.15)

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Type II error

- The probability of not rejecting the null hypothesis H_0 drops as the difference between μ and μ_0 increases or n increases
- 1-$\beta(\mu)$ is called the **power function** of the test. It represents the probability of rejection of H_0 when the mean is μ



$$\beta(\mu) = \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)$$

$$d = \left|\frac{\mu - \mu_0}{\sigma/\sqrt{n}}\right|$$

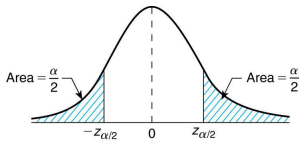
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Choosing size

- What is the sample size n such that
  - the probability of not rejection H_0: μ=μ_0,
  - when the mean is actually μ_1,
  - is equal to some value β?
- We want to find n for which $\beta(\mu_1) = \beta$
  - but cannot solve in closed form for n in
  - $\beta = \Phi\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)$
- An approximate solution is $n \approx \left(\frac{(z_{\alpha/2}+z_\beta)\sigma}{(\mu_1-\mu_0)}\right)^2$

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Example



- An approximate solution is $n \approx \left(\frac{(z_{\alpha/2}+z_\beta)\sigma}{(\mu_1-\mu_0)}\right)^2$
- How many signals should be sent so that at the 95% conf level, the probability of rejection is at least 75%, if the true mean is 9.2? Assume, null hypothesis mean is 8, noise is normal with mean 0 and variance 4
  - $z_\beta = \text{norminv}(0.25) = 0.67$
  - $n \approx (1.96 + 0.67)^2 4/(9.2 - 8.0)^2 = 19.21$
- Hence we must send at least 20 signals. For this value
  - $\beta = \Phi\left(\frac{-1.2}{2/\sqrt{20}} + 1.96\right) - \Phi\left(\frac{-1.2}{2/\sqrt{20}} - 1.96\right) \approx 0.235$
  - We fail to reject 23.5% of the time

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Underspecified parameters

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GoF: Underspecified parameters

- In the dice example, the expected counts were clear. What if we do not have probabilities?
- Example from the book: Consider the number of accidents in an industrial plant over a 30 week period (mean is 95/30 = 3.17)
  - In such examples, we expect the rate λ of accidents to be constant
  - Ross suspects that this data comes from a Poisson distribution
- If the plant experiences an average of say, 3 accidents every 30 weeks, what is the probability that exactly 2 accidents occur during the next 30 weeks?

8	0	0	1	3	4	0	2	12	5
1	8	0	2	0	1	9	3	4	5
3	3	4	7	4	0	1	2	1	2

95 accidents over 30 weeks

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Binomial

8	0	0	1	3	4	0	2	12	5
1	8	0	2	0	1	9	3	4	5
3	3	4	7	4	0	1	2	1	2

- Plant experiences average of, say, 3 accidents/week (over 30 weeks)
- Divide each of the 30 weeks into n very small intervals
  - The probability of an accident in a particular interval is λ/n, where $\lambda = 3$
  - Accidents in different intervals occur independently.
  - The intervals are sufficiently short that the probability of two or more accidents in one interval is negligible
- The number of intervals containing an accident is then approximately

$$X_n \sim \text{Binomial}\left(n, \frac{\lambda}{n}\right)$$



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Poisson

8	0	0	1	3	4	0	2	12	5
1	8	0	2	0	1	9	3	4	5
3	3	4	7	4	0	1	2	1	2

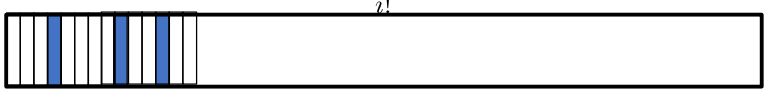
- # of intervals with accident approximately $X_n \sim \text{Bin}\left(n, \frac{\lambda}{n}\right)$
  - As n increases, the probability decreases, but expected number is constant
  - For exactly i accidents

$$P(X_n = i) = \binom{n}{i}\left(\frac{\lambda}{n}\right)^i\left(1-\frac{\lambda}{n}\right)^{n-i}$$

This is a discrete distribution

$$= \underbrace{\frac{n(n-1)\cdots(n-i+1)}{n^i}}_{\longrightarrow 1}\frac{\lambda^i}{i!}\underbrace{\left(1-\frac{\lambda}{n}\right)^n}_{\longrightarrow e^{-\lambda}}\underbrace{\left(1-\frac{\lambda}{n}\right)^{-i}}_{\longrightarrow 1}$$

$$= e^{-\lambda}\frac{\lambda^i}{i!}$$



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GoF: Underspecified parameters

- Consider the number of accidents in an industrial plant over a 30 week period (mean is 3.17)
  - Ross suspects that this data Y comes from a Poisson distribution
- Divide the number of accident into a **table of counts**
  - 0 accidents in a week $\Pr[Y = 0] = e^{-\lambda} = e^{-3.17} = .042$
  - $T \approx 22.1$, df=3, $p \approx 0.00006$ Reject

$$= e^{-\lambda}\frac{\lambda^i}{i!}$$

df = k – 1 – 1 = 3

Accidents k	0	1	2 or 3	4 or 5	> 5
Observed O_i	6	5	8	6	5
Expected $E_i = 30\,p_i$	1.26	4.00	13.03	8.65	3.06
$(O_i - E_i)^2/E_i$	17.83	0.25	1.94	0.81	1.25

8	0	0	1	3	4	0	2	12	5
1	8	0	2	0	1	9	3	4	5
3	3	4	7	4	0	1	2	1	2

In general, df = k–1 –m with m underspecified parameters

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GoF: Illustrative data

- Poisson ($\hat{\lambda} = 3.0$) fit: $T \approx 2.73$, df=3, $p \approx 0.43$. Do not reject H_0
- For a Poisson distribution, mean = variance = λ, MLE = sample mean

Accidents k	0	1	2 or 3	4 or 5	> 5
Observed O_i	3	4	12	7	4
Expected $E_i = 30\,p_i$	1.49	4.48	13.44	8.06	2.52
$(O_i - E_i)^2/E_i$	1.52	0.05	0.15	0.14	0.87

2	4	1	3	0	2	5	3	7	2
1	3	6	2	4	0	2	5	3	4
2	7	1	4	3	2	5	0	6	1

90 accidents over 30 weeks

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